x	−2	−1	0	1	2
$2x$	−4	−2	0	2	4
1	1	1	1	1	1
$y = 2x+1$	−3	−1	1	3	5

Points to be plotted :

(−2, −3), (−1, −1), (0,1), (1,3), (2, 5)

x	−2	−1	0	1	2
$2x$	−4	−2	0	2	4
1	1	1	1	1	1
$y = 2x+1$	−3	−1	1	3	5

Points to be plotted :

(−2, −3), (−1, −1), (0, 1), (1, 3), (2, 5)

Here both the equations are identical; they were only represented in different forms. Since they are identical, their solutions are same. All the points on one line are also on the other!

This means we have an infinite number of solutions which are the ordered pairs of all the points on the line.

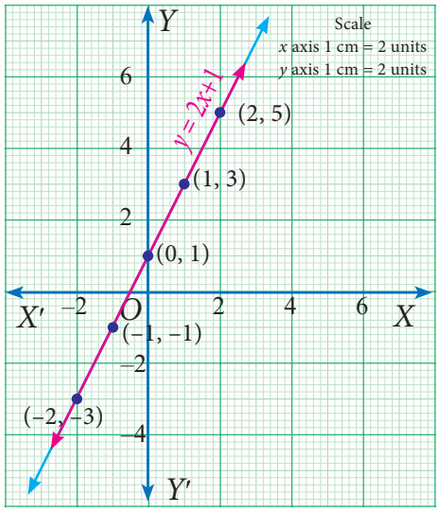


Fig. 3.23

Example 3.47

The perimeter of a rectangle is 36 metres and the length is 2 metres more than three times the width. Find the dimension of rectangle by using the method of graph.

Solution

Let us form equations for the given statement.

Let us consider l and b as the length and breadth of the rectangle respectively.

Now let us frame the equation for the first statement

Perimeter of rectangle = 36

$$2(l + b) = 36$$

$$l + b = \frac{36}{2}$$

$$l = 18 - b \quad \ldots (1)$$

b	2	4	5	8
18	18	18	18	18
−b	−2	−4	−5	−8
$l = 18-b$	16	14	13	10

Points: (2,16), (4,14), (5,13), (8,10)

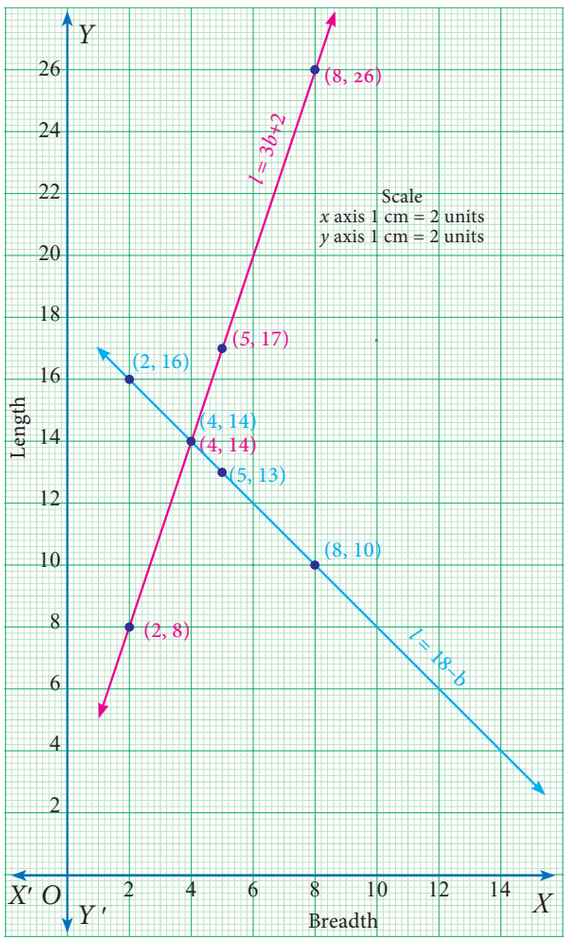


Fig. 3.24

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